



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Automated Data Cleaning Pipeline	
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Abstract

Data is an essential resource for data analysis and machine learning, but real-world datasets often contain missing values, duplicate records, outliers, inconsistent formats, and noisy information. These data-quality issues can affect the accuracy and reliability of analysis and may reduce the performance of machine learning models. Manual identification and correction of such problems can also be time-consuming and repetitive. To address these challenges, this project proposes an **Automated Data Cleaning Pipeline** that automatically detects and handles common data-quality problems in raw datasets. The system accepts a dataset as input and performs data inspection, preprocessing, and cleaning operations. It identifies missing values and applies suitable imputation techniques, detects and removes duplicate records, identifies potential outliers in numerical attributes, and handles inconsistent or improperly formatted data. The pipeline also provides a comparison of the dataset before and after cleaning to help users understand the improvements made during preprocessing. An interactive dashboard is incorporated to display dataset information, detected issues, cleaning results, and visualizations in an easy-to-understand manner. The cleaned dataset can then be exported for further data analysis and machine learning applications. The proposed system aims to reduce manual preprocessing effort, improve data consistency and reliability, and provide an efficient and user-friendly approach for preparing raw datasets for data-driven applications.

Furthermore, the pipeline is designed to make the data-cleaning process more systematic by applying a sequence of preprocessing steps to the input dataset. Each stage of the pipeline focuses on a specific data-quality issue and contributes to improving the overall structure of the dataset. The system can generate useful summaries of the detected problems, allowing users to identify which columns require greater attention during the cleaning process. This approach helps maintain consistency across different datasets and reduces the need for repeated manual inspection.

The proposed pipeline can be applied to datasets from different domains, making it a reusable solution for various data-driven applications. By combining automated preprocessing with an interactive visualization interface, the system provides users with both cleaned data and a clear understanding of the transformations performed. The final output is a structured and analysis-ready dataset that can be utilized for exploratory data analysis, visualization, statistical operations, and machine learning workflows. Overall, the project demonstrates how automation can simplify the data preparation stage, enhance data quality, and establish a reliable foundation for subsequent data science tasks.